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Python Programming Foundations

Week 1 – First Programs and Basic Values

Day 1: "What a Python Program Does"

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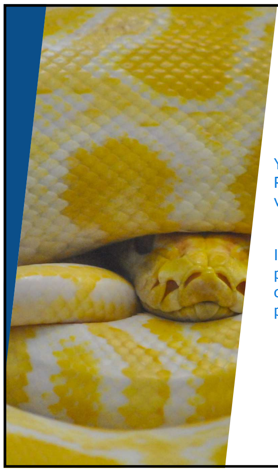

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Make Something Happen

Your first goal is to make Python do something visible.

If you can run a tiny program and explain the output, you are doing real programming.

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First Program

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Start Small, Build Outward

1

You are not expected to know everything today; you are expected to make one

This course starts with small working programs and builds outward.

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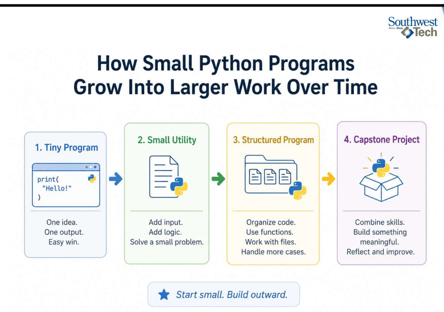

Program Growth

5

What Counts as Success Today

- run the program
- change one value
- run it again
- explain what changed

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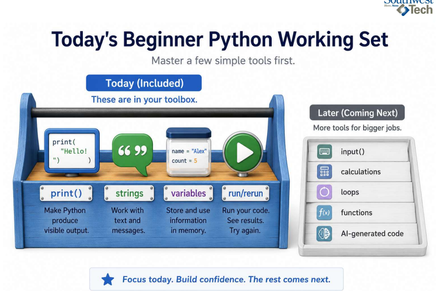
What We Will Use Today

- `print()`
- strings
- variables
- visible output

7



Working Set

8

What We Will “Skip” For Now

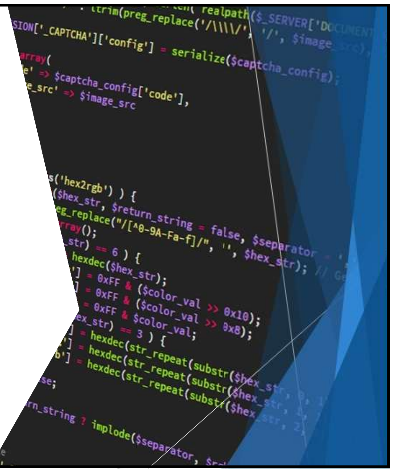
- `input()`
- Calculations
- Loops
- Functions
- AI-generated code

9

A Program Is Instructions

Python follows the instructions in your file.

Most beginner programs run from top to bottom - one line at a time.



10

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Python Runs Instructions Line by Line

1 `course_name = "Python Programming"`

2 `student_name = "Jordan"`

3 `print(student_name)`

Python runs the next instruction in order.

Line-by-Line Exectuion

11

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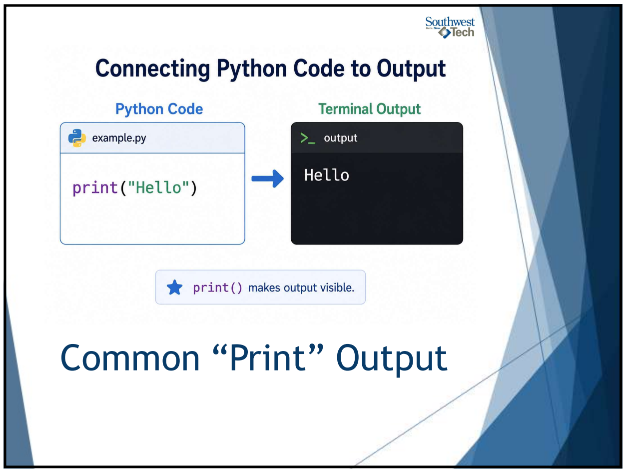
\`print()\` Makes Output Visible

\`print()\` makes a value or message appear on the screen.

Visible output is your first evidence that the program ran.

12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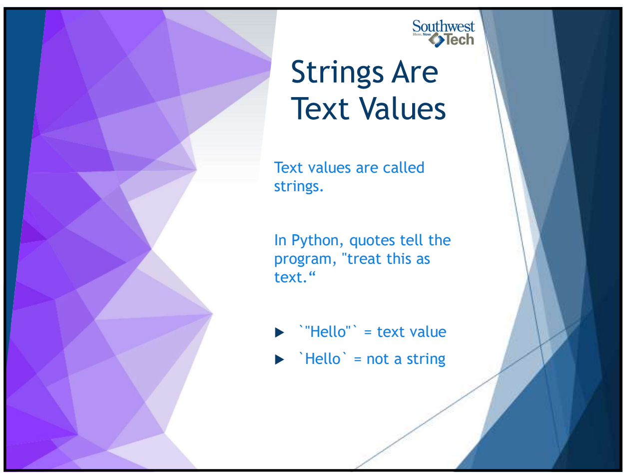
Connecting Python Code to Output
Python Code
example.py
print("Hello")
Terminal Output
>_ output
Hello
print() makes output visible.
Common “Print” Output

13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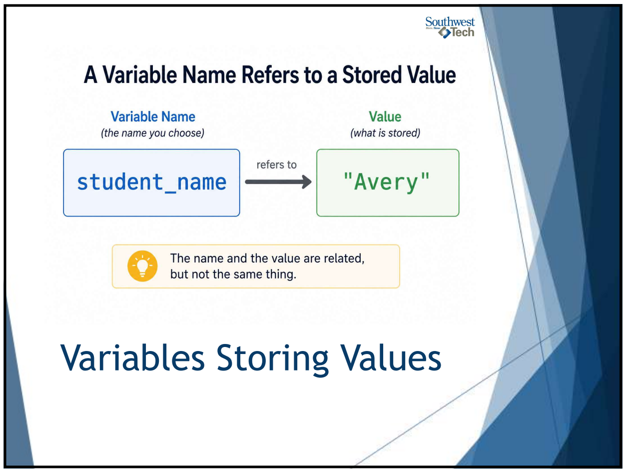
Strings Are Text Values
Text values are called strings.
In Python, quotes tell the program, "treat this as text."
`"Hello"` = text value
`Hello` = not a string

14


Variables Store Values
A variable is a name that refers to a stored value.
The name and the value are related, but they are not the same thing.

15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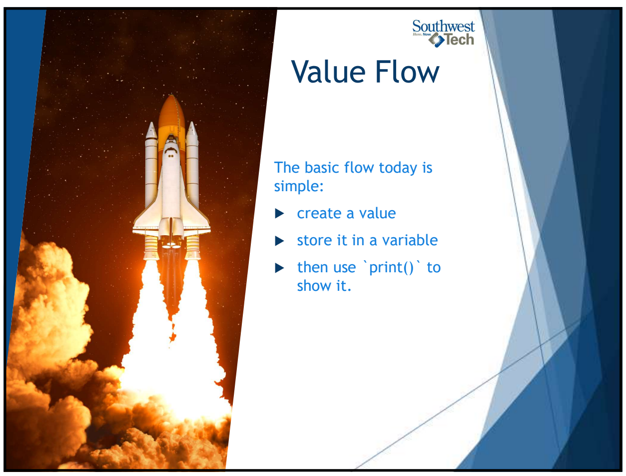
A Variable Name Refers to a Stored Value
Variable Name
(the name you choose)
Value
(what is stored)
student_name
refers to
"Avery"
The name and the value are related, but not the same thing.
Variables Storing Values

16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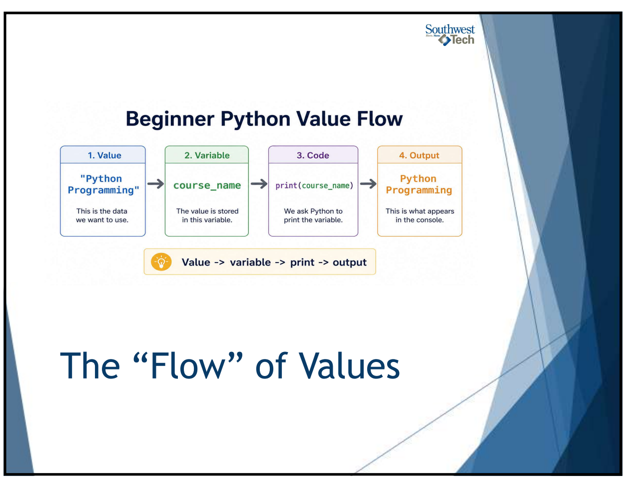
Value Flow
The basic flow today is simple:
create a value
store it in a variable
then use `print()` to show it.

17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Beginner Python Value Flow
1. Value
"Python Programming"
This is the data we want to use.
2. Variable
course_name
The value is stored in this variable.
3. Code
print(course_name)
We ask Python to print the variable.
4. Output
Python Programming
This is what appears in the console.
Value -> variable -> print -> output
The "Flow" of Values

18

Demo #1
“Print String Variables”

19

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What to watch for
As the demo runs, watch for two things:
which values are stored
which lines make output appear

20

Demo #2
“Strings & Variable Updates”

21


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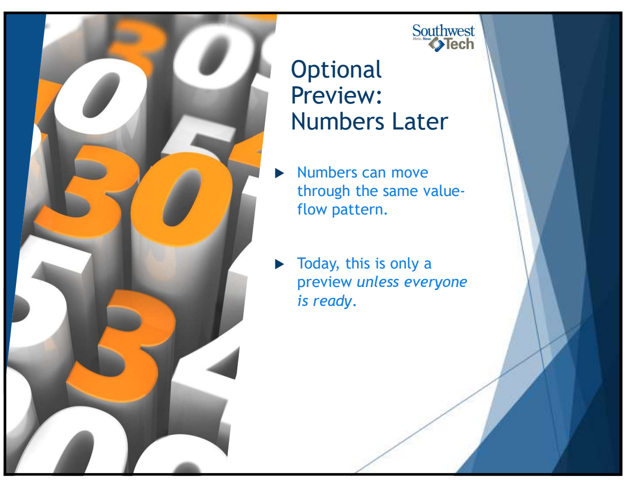
What to watch for
As the demo runs, watch for two things:
When a variable is assigned a new value …
later output can change.

22


Optional Demo #3
“Numbers Expressions”

23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Optional Preview: Numbers Later
Numbers can move through the same value-flow pattern.
Today, this is only a preview *unless everyone is ready*.

24

Common Failure: Missing Quotes

`"Hello"` is a string because it has quotes.

`Hello` without quotes is not automatically text to Python.

25

Why Quotes Matter for Text in Python

Quotes tell Python this is text.

Text Value	Not Text Yet
“Hello”	Hello
quotes mark text	Python reads this differently

Use quotes when you want to work with text.

Quotes are *Important*

26

Common Failure: Name vs Value

The variable name is how we refer to the value.

Changing the stored value can change what the program prints later.

27


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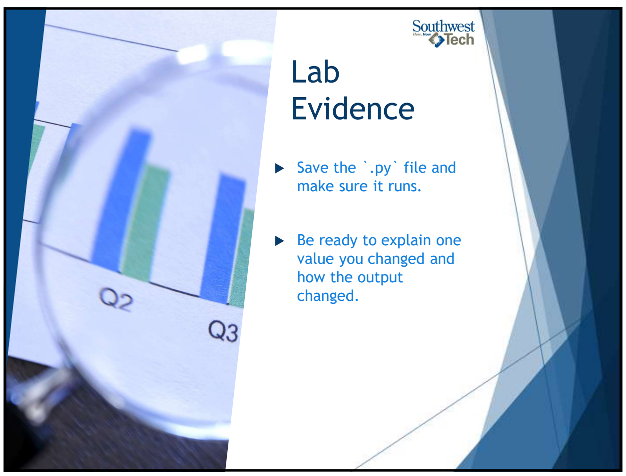
From Demo to Lab
In Lab/Assignment 01:
Create one tiny program with a greeting and one variable. Then change the variable value and rerun the program.
1. Create a new `.py` file.
2. Print a greeting.
3. Store one text value in a variable.
4. Print that variable.
5. Change the stored value.
6. Rerun the program.

28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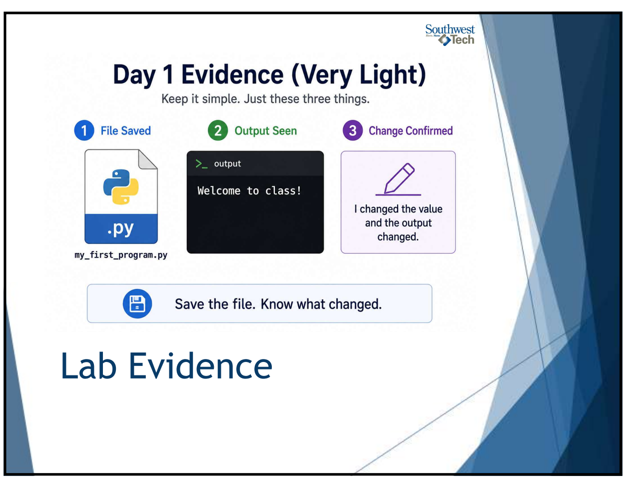
Lab Evidence
Save the `.py` file and make sure it runs.
Be ready to explain one value you changed and how the output changed.
Q2
Q3

29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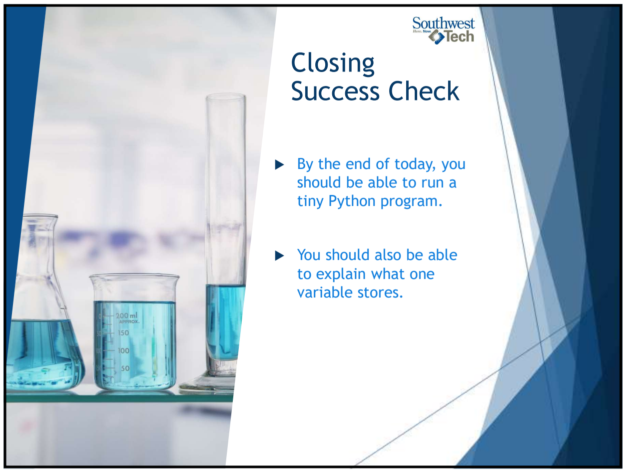
Day 1 Evidence (Very Light)
Keep it simple. Just these three things.
1 File Saved
2 Output Seen
3 Change Confirmed
.py
my_first_program.py
>_ output
Welcome to class!
I changed the value and the output changed.
Save the file. Know what changed.
Lab Evidence

30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Closing
Success Check
By the end of today, you should be able to run a tiny Python program.
You should also be able to explain what one variable stores.

31


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Thank you!
Have Fun!

32